The Journal of Open Source Software

gaia: An R package to estimate crop yield responses to temperature and precipitation

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 14 October 2024
Published: unpublished

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Summary

gaia is an open-source R package designed to estimate crop yield shocks in response to annual weather variations and CO_2 concentrations at the country scale for 12 major crops. It enables the projection of annual yield shocks under various future climate scenarios, differentiated by crop type, country, and year. This innovative tool streamlines the workflow from raw climate data processing to projections of annual shocks to crop yields at the country level, using the response surfaces developed and documented in Waldhoff et al. (2020), an empirical econometric model that leverages historical weather, CO_2, and crop yield data for robust empirical fitting for 12 crops. gaia uses these response surfaces with bias-corrected, gridded monthly temperature and precipitation projections (e.g., from the Coupled Model Intercomparison Project Phase 6 (CMIP6, O'Neill et al. (2016)) and Inter-Sectoral Impact Model Intercomparison Project (ISIMIP, Warszawski et al. (2014))) to project shocks that can be applied to agricultural productivity changes at the country-level for use in economic models. The historical and future projections use gridded, country-crop specific monthly growing season precipitation and temperature data, aggregated to the national level, weighted by cropland area derived from MIRCA (Portmann et al., 2010). These annual, country, crop-specific yield shocks can be aggregated to different regional definitions, crop commodity definitions, and time periods (e.g., 20-year rolling-average trends), as needed by specific economic models. gaia serves as a lightweight, powerful model that equips researchers with projections of annual yield shocks for multiple crops, at a model-specific spatial resolution that is necessary to explore crop yields responses to a broad range of future climate projections, enhancing human-Earth system analysis capabilities.

Statement of need

Agricultural production is highly responsive to annual weather patterns, which are becoming increasingly variable in response to climate change (e.g. Iizumi & Ramankutty (2016)). Agricultural markets are global, with significant volumes of international trade, and yield shocks due to this variability in one region will therefore impact others. In addition, agricultural production is closely connected with energy and water sectors. Improving the representation of agricultural production in global multi-sector dynamics models is the key to better understanding economic impacts of climate change that account for human-Earth system interactions. To better represent future agricultural production and study these interactions, multisector dynamics models, such as the Global Change Analysis Model (GCAM, Calvin et al. (2019), Joint Global Change Research Institute (2023)), require projections of yield responses to changing weather patterns, which are key drivers of production and vital to understanding the co-evolution of energy, water, and land systems.

Most economic, multisectoral modeling of climate change impacts on agriculture has relied on the representation of the effects of long term, gradual trends in temperature and precipitation on crop yields for a few key crops (Lampe et al., 2014). Improved understanding of the effects of climate change on agricultural production requires projections of the impacts of annual weather changes on a broad range of crops for a range of climate projections (Ray et al., 2015). However, the current crop modeling landscape lacks globally comprehensive, computationally efficient, validated models capable of simulating yield responses to both inter-annual variability and long-term trends across a wide range of climate change scenarios and crop types (Waldhoff et al., 2020). To address the gap, we have developed `gaia`, which uses the empirical model described in Waldhoff et al. (2020). `gaia` produces projections of both future annual variations and long-term trends in yield changes at the country-level for a wide range of crops (cassava, cotton, maize, rice, potatoes, sorghum, soybean, sugar beet, sugarcane, sunflower, and wheat) using gridded monthly temperature and precipitation inputs for any future climate scenario.

States of the Field

The exploration of effects of future climate change on agricultural production is crucial for providing insights into global food, energy, water, and economic development. Various crop models, including process-based and empirical models, have been developed to simulate crop yields under different climate scenarios (Chapagain et al., 2022; Rauff & Bello, 2015). Müller et al. (2017) conducted a global gridded crop model (GGCM) intercomparison experiment involving 14 process-based crop models, which allow users to simulate diverse crop management options, soil types, and weather conditions. However, GGCMs require large amounts of site-specific data for calibration, making extension to many crops and regions more challenging (Müller et al., 2017). In addition, these models can take significant time to run and the yield shock projections are often available for only a limited number of climate change scenarios.

These considerations may limit the usefulness of yield changes from process models as inputs to global economic, multisector models, which include a wider range of crops and many future climate scenarios. In the multisector dynamics modeling field, there is a need for projections of future crop yield shocks due to changing climate and weather patterns that can strike a balance between simulation fidelity and lightweight modeling capabilities. Such models are essential for exploring a wide range of future climate scenarios with global coverage. `gaia` achieves this balance by employing computationally efficient statistical methods to model 12 major crops globally, while maintaining the sensitivity to capture both gradual climate changes and interannual variability impacts on crop yields. In addition, users can estimate yield shocks for additional crops by providing the appropriate input data.

Functionality

`gaia` is designed as an empirical model, which is driven by monthly average growing season temperature and precipitation, integrated into an efficient modular structure. This architecture streamlines the entire workflow, from initial climate model and crop yield data inputs and processing, through empirical model fitting, yield shock projections under future climate scenarios, to the calculation of agricultural productivity changes for GCAM. The modular design also facilitates comprehensive diagnostic outputs, enhancing the tool's utility for researchers and decision makers.

The primary functionality of `gaia` is encapsulated in the `yield_impact` wrapper function, which executes the entire workflow from climate data processing to yield shock estimation. Users can also execute individual functions to work through the main steps of the process (Figure 1). Detailed instructions on `gaia` can be accessed at https://jgcri.github.io/gaia.

1. `weighted_climate`: Processes CMIP-ISIMIP climate NetCDF data and calculates cropland-weighted precipitation and temperature at the country level, differentiated by

crop type and irrigation type. The function accepts both daily or monthly climate data that are consistent with the CMIP-ISIMIP NetCDF data format

2. `crop_calenders`: Generates crop planting months for each country and crop based on crop calendar data (Sacks et al., 2010).
3. `data_aggregation`: Calculates crop growing seasons using climate variables processed by `weighted_climate` and crop calendars for both historical and projected periods. This function prepares climate and yield data for subsequent model fitting.
4. `yield_regression`: Performs regression analysis fitted with historical annual crop yields, monthly growing season temperature and precipitation, CO_2 concentrations, GDP per capita, and year. The default econometric model applied in `gaia` is from Waldhoff et al. (2020). User can specify alternative formulas that are consistent with the data processed in `data_aggregation`.
5. `yield_shock_projection`: Projects yield shocks for future climate scenarios using the fitted model and temperature, precipitation, and CO_2 projections from the climate scenario.
6. `gcam_agprodchange`: Remaps country-level yield shocks to GCAM-required spatial scales (i.e., region, basin, technology intersections), based on harvested areas, and aggregates crops to GCAM commodities. This function applies the projected shocks to GCAM scenario agricultural productivity growth rates (the unit used to project future yields in GCAM) and creates ready-to-use XML outputs for GCAM.

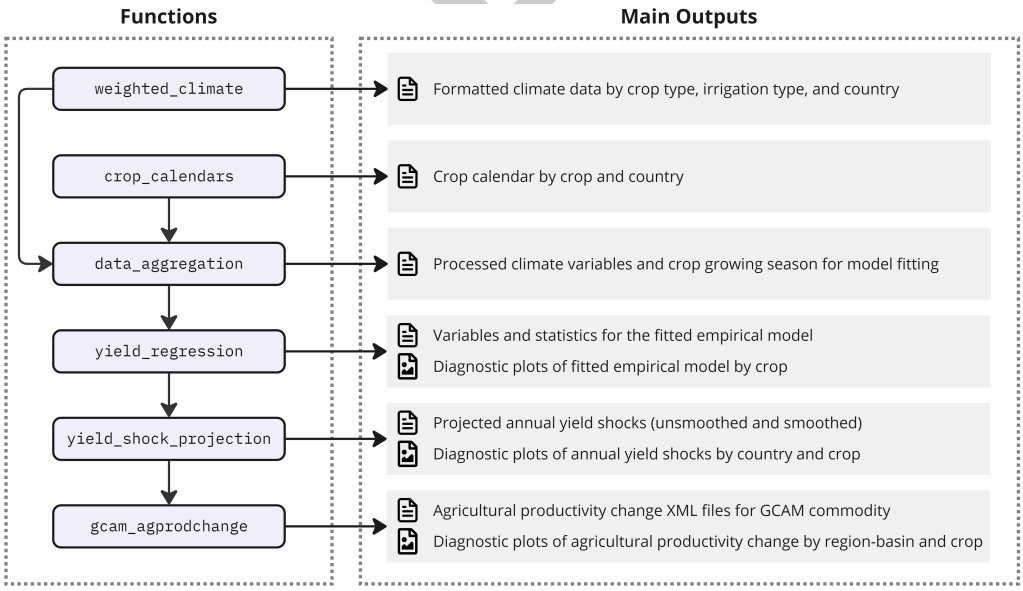

Figure 1: The gaia workflow showing the functions and the corresponding outputs of modeling crop yield shocks to weather variables using an empirical econometric model.

Acknowledgements

This research was supported by the US Department of Energy, Office of Science, as part of research in MultiSector Dynamics, Earth and Environmental System Modeling Program. The Pacific Northwest National Laboratory is operated for DOE by Battelle Memorial Institute under contract DE-AC05-76RL01830. The views and opinions expressed in this paper are those of the authors alone.

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